Yassine Ben Maktouf

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Junior Actuary and Data Science Engineer

Experience

- **Actuarial Data Scientist (End-of-Studies Internship) | Vermeg, Tunis** *Feb 2026 - Jul 2026*
  - Worked on PERI retirement-savings products in Solife, Vermeg's life-insurance and pensions administration platform: automated product configuration from specification documents, with a deterministic, auditable engine that applies every change to regulated product rules (XML, Groovy).
  - Built a search assistant over an actuarial and insurance document corpus (hybrid retrieval, RAG) and evaluated it quantitatively (MRR, Recall, NDCG) to select the production configuration.
  - Ran evaluation studies comparing extraction models (fine-tuned XLM-RoBERTa vs. LLMs) and auditing answer faithfulness, so that results stay traceable in a regulated insurance setting.
- **Data Science Intern | Les Laboratoires MédiS, Nabeul** *Jun 2025 - Aug 2025*
  - Built a sales forecasting model, comparing machine learning and deep learning approaches and selecting the most accurate.
  - Automated the data pipeline on SQL Server and delivered interactive reporting dashboards for decision-makers.
- **Software Engineering Intern | STIP, Tunis** *Jun 2024 - Jul 2024*
  - Built a SQL Server database, Node.js REST APIs and a React interface to manage employees, phone numbers and contracts, with role-based access control.

Key Projects

- **Actuarial Portfolio Heterogeneity Study (R):** Calibrated the Lee-Carter model on French mortality of smokers and non-smokers (1950 cohort), projected future mortality trends, and designed fair annuity pricing for each risk group.
- **Health-Insurance Claims Platform:** Automated the processing of handwritten medical claim documents (French/Arabic) with OCR and document classification, added signature-based fraud detection (Siamese network) and estimated reimbursements against official reference databases. Presented at ESPRIT's 12th Bal des Projets (2025).
- **Customer Churn Modelling:** Classification models (Random Forest, Decision Tree) on SMOTE-balanced data and PCA/K-Means segmentation to identify churn drivers, the same techniques used for lapse and retention analysis.
- **Hackathon Projects:** Imaginify AI, an AI story-generation web app (1st place, Special AI Models Hack, lablab.ai, 2024); FinAssist, an AI personal-finance assistant that analyses Excel data (2nd place, ChatHack, lablab.ai, 2023); NASA International Space Apps Challenge (2024).

Education

- **Master's in Actuarial Science (double degree) | IRA**, Le Mans University, France *2024 - 2026*
  Courses: life and non-life actuarial science, risk theory, reinsurance and extreme risks, insurance regulation, accounting and prudential standards, stochastic models for actuarial science and finance, mathematical models for finance, Monte Carlo methods, advanced numerical methods, non-life reserving (Chain Ladder, Mack, Bootstrap), pricing (GLM).
- **Engineering Degree in Computer Science, Data Science Specialty | ESPRIT**, Tunisia *2023 - 2026*
  Courses: probability and statistics, time series, operations research (linear programming, graphs), machine learning, big data, business intelligence.
- **Preparatory Classes in Mathematics and Physics | IPEIT**, Tunisia *2021 - 2023*
  Courses: analysis, linear algebra, probability, physics.
- **Baccalaureate in Mathematics | Lycée Ibnou Rachik**, Tunisia *2021*

Skills & Certifications

Actuarial: life and non-life actuarial science, mortality modelling (Lee-Carter), annuity pricing, risk theory, reinsurance and extreme risks, prudential regulation (Solvency II), IFRS 9, IFRS 17, stochastic models, Monte Carlo simulation, Solife (life and pensions platform)
Statistics & machine learning: time-series forecasting, classification, clustering, PCA, model evaluation, scikit-learn, PyTorch, NLP, LLM
Programming & data: R, Python, SQL (SQL Server, MySQL), vector databases, HTML, CSS, Docker
Certifications: Machine Learning with Python (IBM); Applications of AI for Anomaly Detection (NVIDIA); Evaluation and Light Customization of LLMs (NVIDIA); AWS Academy Cloud Foundations
Activities: General Secretary of the ATAST club at ESPRIT (2023 - 2024), club ranked 2nd at the annual BTech competition; Member of the IEEE RAS (Robotics and Automation Society) student chapter at ESPRIT (2023 - 2024)
Languages: Arabic (native), English (bilingual), French (professional)